

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 1: General Performance, Atmosphere (ISA), Wind & Runway

Syllabus Source: CAE Oxford (pp. 1-44) | Pass Target: >= 90% AviationExam

1. ATMOSPHERIC PROPERTIES & ALTITUDE CALCULATIONS

Aircraft performance is governed by air density, which dictates aerodynamic lift and engine thrust output:

- Pressure Altitude (PA): Altitude above the standard 1013.25 hPa datum plane: Pressure Alt = Elevation + 30 ft x (1013.25 - QNH) (or 27 ft/hPa).
- Density Altitude (DA): Pressure altitude corrected for non-standard temperature: Density Alt = Pressure Alt + 120 ft x (OAT - ISA Temperature). High density altitude severely degrades lift, climb rate, and engine thrust!
- ISA Temperature Formula: ISA Temperature (°C) at altitude h (thousands of ft) = +15°C - 2°C x (h / 1,000 ft) (up to tropopause at 36,090 ft / -56.5°C).

2. EASA REGULATORY FACTORED WIND RULES

When taking off or landing, wind provides aerodynamic assistance or penalty. Under EASA regulations, safety margins are legally mandated:

- Headwind Credit (50% Rule): Operators are permitted to take credit for NO MORE THAN 50% of the reported headwind component! (Factored Headwind = 0.50 x Reported Headwind).
- Tailwind Penalty (150% Rule): Operators MUST PENALIZE performance by AT LEAST 150% of the reported tailwind component! (Factored Tailwind = 1.50 x Reported Tailwind).
- Maximum Certified Tailwind: Standard certified tailwind limit for commercial transport jets is 10 knots (or 15 knots if specifically certified).

3. RUNWAY CONTAMINATION & HYDROPLANING PHYSICS

Runway surface conditions drastically alter friction coefficients and wheel braking authority:

- Dry Runway: Free of visible moisture; maximum braking coefficient.
- Damp / Wet Runway: Moisture present, water depth <= 3 mm. Reduces friction coefficient; landing distance factored by x 1.15.
- Contaminated Runway: More than 25% of the required runway surface is covered with standing water, slush, or loose snow exceeding 3 mm depth (or compacted snow / ice).
- Dynamic Hydroplaning Formula: Speed at which tires completely ride on water film with zero tire-to-pavement friction: $V_p \text{ (kt)} = 9 \times \sqrt{P}$ [P in psi], or $V_p \text{ (kt)} = 5.7 \times \sqrt{P}$ [P in bar].

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Density Altitude & Hydroplaning

Problem 1 (Density Altitude): Field elevation 4,000 ft, QNH 1003 hPa, OAT +25 C. -

Step 1: Pressure Altitude PA = $4,000 + 30 \times (1013 - 1003) = 4,300$ ft. - Step 2: ISA

Temp at 4,000 ft = $15 - (2 \times 4) = +7$ C. - Step 3: ISA Deviation = $+25 - (+7) = +18$

C. - Step 4: Density Altitude DA = $4,300 + 120 \times 18 = 6,460$ ft. Problem 2

(Hydroplaning Speed): Tire pressure 100 psi. - Formula: $V_{\text{hydro}} = 9 \times \sqrt{100} = 9 \times$

$10 = 90$ knots. - Above 90 kt on standing water (>3 mm), tire completely loses braking

friction!

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)**[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Factored Wind & Dynamic Hydroplaning Speed**

SCENARIO (Standard AviationExam Drill): A twin-jet is preparing for landing on Runway

27 (magnetic heading 270°): - Tower reported wind: 330° at 20 knots - Main gear tire

pressure: 144 psi QUESTION 1: What is the FACTORED HEADWIND component for performance

calculations? QUESTION 2: At what ground speed will dynamic hydroplaning occur if

runway has standing water? SOLUTION PASO A PASO (FOR DUMMIES): Part 1: Calculate

Reported Wind Components: - Wind angle to runway = $330^\circ - 270^\circ = 60^\circ$ off runway

heading. - Reported Headwind = Wind Speed $\times \cos(60^\circ) = 20 \text{ kt} \times 0.50 = 10.0$ knots.

- Reported Crosswind = Wind Speed $\times \sin(60^\circ) = 20 \text{ kt} \times 0.866 = 17.3$ knots. Part 2:

Apply EASA 50% Headwind Safety Factor: - Factored Headwind = 50% of Reported

Headwind = $0.50 \times 10.0 \text{ kt} = 5.0$ knots! - (You are ONLY allowed to credit 5 knots of

headwind assistance in your landing distance calculation!). Part 3: Calculate Dynamic

Hydroplaning Speed (V_p): - Formula: $V_p = 9 \times \sqrt{\text{Tire Pressure in psi}}$ - Tire

Pressure = 144 psi. $\sqrt{144} = 12$. - $V_p = 9 \times 12 = 108$ knots! FINAL ANSWER:

Factored Headwind = 5.0 knots. Dynamic hydroplaning occurs above 108 knots!

[!] EXAM TRAP (AVIATIONEXAM) - Headwind vs Tailwind Factoring Trap

- Headwind: CREDIT ONLY 50% (Multiply by 0.50). Gives a pessimistic, safe distance. -

Tailwind: PENALIZE BY 150% (Multiply by 1.50). Gives an extra-long, safe stopping

distance. - Never mix these up! Multiplying tailwind by 0.50 would be a fatal safety error!

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA regulations, what proportion of the reported HEADWIND component may be credited for take-off?

- [A]: 100%
- [B]: Not more than 50%
- [C]: 150%
- [D]: Zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. EASA CS-25 requires that no more than 50% of nominal headwind component be used in performance graphs, providing a safety buffer against wind lulls.

Q2: An aircraft's main tires are inflated to 100 psi. What is the approximate dynamic hydroplaning speed on a runway with standing water?

- [A]: 100 knots
- [B]: 90 knots

- [C]: 60 knots
- [D]: 140 knots

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Formula: $V_p = 9 \times \sqrt{P} = 9 \times \sqrt{100} = 9 \times 10 = 90$ knots. Above 90 kt, a water wedge completely lifts the tire off the tarmac.

Q3: How does an UPHILL runway slope affect take-off performance and landing performance?

- [A]: Improves take-off and degrades landing
- [B]: Increases take-off distance (gravity opposes acceleration) and DECREASES landing distance (gravity assists deceleration)
- [C]: Has zero effect
- [D]: Decreases both distances

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. An uphill slope acts as a retardant: it lengthens take-off ground run because gravity opposes engine thrust, but shortens landing roll because gravity helps decelerate the aircraft.

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 2: Single-Engine Piston Aircraft Performance (Class B)

Syllabus Source: CAE Oxford (pp. 45-78) | Pass Target: $\geq 90\%$ AviationExam

1. CLASS B AIRCRAFT CERTIFICATION (CS-23)

Class B covers propeller-driven aircraft with a Maximum Certified Take-off Mass of 5,700 kg or less, and 9 passenger seats or fewer:

- Take-Off Distance (TOD to 50 ft): The horizontal distance from brake release to the point where the aircraft reaches a screen height of 50 ft above runway surface, with all engines operating.
- Best Angle of Climb Speed (V_x): The speed providing the maximum gain of altitude per unit horizontal distance (steepest climb gradient). Used for clearing close-in obstacles after take-off!
- Best Rate of Climb Speed (V_y): The speed providing the maximum gain of altitude per unit time (highest ft/min). Used for normal en-route climb to cruising altitude.
- Absolute Ceiling vs Service Ceiling: Absolute Ceiling is where maximum rate of climb drops to zero ($V_x = V_y$). Service Ceiling for single-engine aircraft is defined as the pressure altitude where maximum climb rate drops to 100 ft/min.

2. CLIMB GRADIENT VS RATE OF CLIMB FORMULAS

- Climb Gradient (%): $\text{Gradient (\%)} = (\text{Vertical Rise} / \text{Horizontal Distance}) \times 100 = (\text{Thrust} - \text{Drag}) / \text{Weight} \times 100$
- Rate of Climb (ROC) Master Rule: $\text{ROC (ft/min)} = \text{Gradient (\%)} \times \text{TAS (kt)}$ (approximate rule of thumb: $\text{ROC} \sim \text{Gradient \%} \times \text{True Airspeed}$).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Class B Single-Engine Factored Distance

Problem: POH un-factored takeoff distance to 50 ft screen height is 400 m on dry paved runway. Step 1: Apply EASA commercial safety factor ($\times 1.25$): - Required Distance = $400 \text{ m} \times 1.25 = 500 \text{ meters}$. Step 2: Dry grass correction (+10%): $500 \text{ m} \times 1.10 = 550 \text{ meters}$. Step 3: Uphill runway slope correction (+2% slope adds approx +10%): $550 \text{ m} \times 1.10 = 605 \text{ meters}$. Step 4: Takeoff is legal only if TORA $\geq$ required factored distance!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Clearing an Obstacle on Take-Off

SCENARIO (Single-Engine Take-off Problem): A single-engine aircraft takes off from a runway: - Take-Off Distance to 50 ft screen height = 600 meters - An obstacle (tree line) of height 110 ft (33.5 m) is located 1,600 meters from brake release - The aircraft climbs at $V_x = 70$ kt TAS with an average climb gradient of 8.0% QUESTION:

Will the aircraft clear the obstacle, and what will be its vertical clearance margin?

SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Determine the horizontal climb distance from 50 ft screen height to the obstacle: - Obstacle distance from brake release = 1,600 m - Distance to screen height (50 ft) = 600 m - Climb distance = 1,600 m - 600 m = 1,000 meters. Step 2: Calculate the altitude gained over this 1,000 m climb distance: - Climb Gradient = 8.0% = 0.08 - Altitude gained = Distance x Gradient = 1,000 m x 0.08 = 80 meters (approx 262 ft). Step 3: Add the initial screen height (50 ft = 15.2 m) to get total aircraft height: - Total height over obstacle = 15.2 m + 80.0 m = 95.2 meters (312 ft). Step 4: Compare with obstacle height: - Obstacle height = 33.5 m (110 ft) - Vertical Clearance = 95.2 m - 33.5 m = 61.7 meters (approx 202 ft)! FINAL ANSWER: Yes! The aircraft clears the tree line with a safe vertical margin of 61.7 meters (202 ft).

[!] EXAM TRAP (AVIATIONEXAM) - V_x vs V_y Wind Effect Trap

- Airspeed for V_x and V_y are INDICATED AIRSPEEDS (IAS) and are NOT affected by wind! - Headwind STEEPENS the flight path gradient over the ground (helps obstacle clearance). - Tailwind FLATTENS the flight path gradient over the ground (danger of hitting obstacles)!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the official definition of 'SERVICE CEILING' for a single-engine piston aircraft?**

- [A]: The altitude where climb rate becomes zero
- [B]: The pressure altitude at which the maximum rate of climb drops to 100 ft/min
- [C]: 10,000 ft
- [D]: The altitude where cabin oxygen is required

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under CS-23 certification, the service ceiling for a single-engine aircraft is reached when the maximum full-power rate of climb decreases to 100 ft/min.

Q2: When flying a single-engine aircraft to clear a tall obstacle immediately after take-off, which climb speed M

- [A]: V_y (Best Rate of Climb)
- [B]: V_x (Best Angle of Climb), because it gives the steepest vertical flight path angle per meter of ground distance
- [C]: V_{no}
- [D]: Maximum cruise speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. V_x produces the maximum altitude gain per horizontal distance traveled, maximizing vertical clearance over close-in obstacles.

Q3: What happens to the Best Angle of Climb Speed (V_x) and Best Rate of Climb Speed (V_y) as altitude increases?

- [A]: Vx decreases and Vy increases
- [B]: Vx INCREASES slightly and Vy DECREASES until they meet at the absolute ceiling
- [C]: Both speeds stay perfectly constant
- [D]: Both drop to zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. With increasing altitude, Vx (IAS) increases slightly while Vy (IAS) decreases. The altitude where Vx equals Vy is the Absolute Ceiling (rate of climb = 0).

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 3: Multi-Engine Class B Piston (Take-Off & OEI Climb)

Syllabus Source: CAE Oxford (pp. 79-114) | Pass Target: $\geq 90\%$ AviationExam

1. ASYMMETRIC FLIGHT & LOSS OF 80% CLIMB PERFORMANCE

When an engine fails on a twin-engine piston aircraft, 50% of the total engine power is lost. **HOWEVER, CLIMB PERFORMANCE DROPS BY 80% OR MORE!** This is because climb depends on **EXCESS POWER** above that required for level flight:

- **Excess Power Law:** Power Available drops by 50%, while Power Required **INCREASES** due to rudder deflection drag and windmilling propeller drag. The remaining margin of excess power is cut by 80% to 100%!
- **Windmilling Drag:** An unfeathered windmilling propeller produces massive drag (equivalent to dragging a large flat plate through the air). **IMMEDIATE ACTION:** Feather the dead propeller!
- **5° Bank Rule:** Banking 2° to 5° **TOWARDS THE OPERATING ENGINE** and centering the ball with half-deflection creates a horizontal component of lift that opposes engine yaw, reducing rudder drag and maximizing OEI climb gradient.

2. MULTI-ENGINE SERVICE CEILING & SPEEDS

- **OEI Service Ceiling:** The altitude where the single-engine rate of climb drops to 50 ft/min (with propeller feathered).
- **Vyse (Blue Line Speed):** Best Rate of Climb Speed with One Engine Inoperative. Marked on the airspeed indicator by a prominent **BLUE RADIAL LINE**. The single most important safety speed in twin-engine emergencies!
- **Vmca (Red Line Speed):** Minimum Control Speed in the Air. Marked by a **RED RADIAL LINE**. Airspeed **MUST NEVER** be allowed to drop below Vmca during asymmetric flight!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Multi-Engine Class B Engine Failure

Step 1: Directional Control: Apply full rudder toward live engine and bank up to 5 degrees into live engine. Step 2: Maintain Airspeed: Pitch to hold VYSE (Blue Line on airspeed indicator). Step 3: Cleanup Airframe: Landing gear UP, Flaps UP. Step 4: Identify & Feather: Dead foot, dead engine. Feather inoperative propeller immediately! Rule: If airspeed drops below VMCA (Red Line), reduce live engine throttle immediately to prevent unrecoverable roll!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Determining OEI Drift Down & Terrain Clearance

SCENARIO (Twin-Engine OEI Emergency): A light twin-engine aircraft is cruising at 9,000 ft MSL: - Aircraft All-Engine Service Ceiling: 18,000 ft - Certified One-Engine Inoperative (OEI) Service Ceiling at current mass: 6,500 ft MSL - The left engine fails catastrophically! - Flight path crosses a mountain ridge with elevation 6,800 ft MSL located 20 miles ahead QUESTION: Can the aircraft maintain level flight at 9,000 ft? What will happen, and can it clear the ridge? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Check if 9,000 ft can be sustained: - OEI Service Ceiling = 6,500 ft MSL. - Because current altitude (9,000 ft) is HIGHER than the OEI ceiling (6,500 ft), the aircraft CANNOT sustain level flight! - The aircraft MUST descend (drift down) toward 6,500 ft. Step 2: Compare OEI ceiling with the mountain ridge: - OEI Service Ceiling = 6,500 ft MSL. - Mountain ridge height = 6,800 ft MSL. - Since 6,500 ft < 6,800 ft, the aircraft will eventually settle BELOW the top of the mountain ridge! Step 3: Operational Decision / Emergency Procedure: - Setting blue line speed (Vyse) will only delay descent down to 6,500 ft. - The pilot MUST IMMEDIATELY TURN AWAY from the mountain ridge toward lower terrain or an emergency diversion airport! FINAL ANSWER: Flight at 9,000 ft cannot be maintained. Aircraft will drift down to 6,500 ft, which is below the 6,800 ft ridge. Immediate diversion is mandatory.

[!] EXAM TRAP (AVIATIONEXAM) - The 50% Power vs 80% Climb Trap

- Losing 1 engine on a twin LOSES 50% OF POWER, BUT 80% TO 90% OF CLIMB RATE! - Example: Twin climbing at 1,000 ft/min on 2 engines does NOT climb at 500 ft/min on 1 engine! It will climb at only 100 to 200 ft/min!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: On a multi-engine piston aircraft airspeed indicator, what does the BLUE RADIAL LINE indicate?**

- [A]: Maximum flap extension speed
- [B]: Vyse (Best Rate of Climb Speed with One Engine Inoperative)
- [C]: Minimum control speed ground
- [D]: Never exceed speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The blue line marks Vyse, the airspeed yielding maximum single-engine rate of climb with the critical engine feathered.

Q2: Why does an aircraft lose approximately 80% of its climb performance when one engine fails on a twin-engine aircraft?

- [A]: Half the fuel is lost
- [B]: Climb depends on EXCESS power; after losing 50% power and adding asymmetric drag, the remaining excess power above level flight is tiny
- [C]: The tailplane stalls
- [D]: Generator fails

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Climb requires excess thrust above total drag. At single-engine speed, almost all remaining thrust is consumed just keeping the airplane in level flight, leaving very little excess power for climb.

Q3: Under CS-23, what is the definition of OEI Service Ceiling for multi-engine piston aircraft?

- [A]: Climb rate = 0 ft/min
- [B]: Altitude where maximum rate of climb with one engine feathered drops to 50 ft/min
- [C]: 100 ft/min
- [D]: 15,000 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. For multi-engine Class B aircraft, OEI Service Ceiling is defined at a residual climb rate of 50 ft/min (compared to 100 ft/min for all-engine).

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 4: Class B En-Route & Factorised Landing Performance

Syllabus Source: CAE Oxford (pp. 115-148) | Pass Target: $\geq 90\%$ AviationExam

1. CRUISE PERFORMANCE: RANGE VS ENDURANCE

Cruising efficiency for propeller aircraft depends on matching engine brake horsepower to airframe drag curves:

- Maximum Range Cruise Speed (V_{MRC}): Flown at the speed of minimum total drag (V_{md} / tangent point on drag curve). Provides the maximum nautical miles per kilogram of fuel burned (NMPK).
- Maximum Endurance Speed (V_{min_power}): Flown at the speed of minimum power required ($V_{mp} \sim 0.76 \times V_{md}$). Provides the maximum flight time (hours) in the air per kilogram of fuel burned. Used for holding patterns!

2. CLASS B COMMERCIAL LANDING SAFETY FACTORS

Under EASA AIR-OPS (CAT.POL.A.330), landing distance on a commercial flight must incorporate legal safety factors:

- The 1.43 Landing Factor: The Landing Distance Available (LDA) at the destination must be at least 1.43 times the actual measured landing distance from 50 ft: Factored Landing Distance = Actual Landing Distance $\times$ 1.43.
- Grass Runway Corrections: Dry grass up to 20 cm: Multiply landing distance by 1.20 (+20%). Wet grass: Multiply by 1.30 (+30%).
- Slope Corrections: Downslope runway: Increases landing distance by +5% per 1% downhill slope!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Class B Landing Distance Factor

Problem: Flight manual demonstration distance from 50 ft to full stop is 500 m.

Landing Distance Available (LDA) is 850 m. Step 1: Calculate Dry Runway Required

Landing Distance (RLD): - $RLD_{dry} = 500 \text{ m} \times 1.43 = 715 \text{ meters}$ (must stop within 70%

of LDA: $1/0.70 = 1.43$). - Check: $715 \text{ m} \leq 850 \text{ m} \rightarrow$ Dry landing is LEGAL! Step 2:

Calculate Wet Runway Required Landing Distance: - $RLD_{wet} = RLD_{dry} \times 1.15 = 715 \text{ m} \times$

$1.15 = 822.25 \text{ meters}$. - Check: $822.25 \text{ m} \leq 850 \text{ m} \rightarrow$ Wet landing is LEGAL!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Factored Landing Distance on Wet Grass Runway

SCENARIO (Commercial Class B Landing Problem): A twin-propeller air-taxi is planning to land at an unpaved airfield: - Unfactored (Flight Manual) Landing Distance from 50 ft on dry asphalt = 500 meters - Destination runway: Wet grass surface (+30% correction) - Runway slope: 2.0% downhill slope (+5% per 1% slope = +10% correction) - EASA commercial operation factor: 1.43 QUESTION: What is the MINIMUM Landing Distance Available (LDA) required to land legally? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Apply the Runway Surface Correction (Wet Grass = x 1.30): - Distance after grass factor = $500 \text{ m} \times 1.30 = 650 \text{ meters}$. Step 2: Apply the Downslope Correction (2% downhill x 5% = +10% -> x 1.10): - Distance after slope factor = $650 \text{ m} \times 1.10 = 715 \text{ meters}$. Step 3: Apply the mandatory EASA Commercial Safety Factor (x 1.43): - Factored Required LDA = $715 \text{ m} \times 1.43 = 1,022.45 \text{ meters}$ -> 1,023 meters! FINAL ANSWER: The runway must have a certified Landing Distance Available (LDA) of AT LEAST 1,023 meters! (More than double the basic 500 m manual distance!).

[!] EXAM TRAP (AVIATIONEXAM) - The Sequential Multiplication Trap

- Always apply factors sequentially (Surface -> Slope -> Regulatory factor). - Notice how a 500 m basic distance turns into 1,023 m! In ATPL exams, failing to apply the 1.43 commercial factor is the most common student error!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA regulations, what is the regulatory multiplication factor applied to the landing distance of a C

- [A]: 1.15
- [B]: 1.43 (meaning required LDA must be at least 143% of actual distance)
- [C]: 1.67
- [D]: 2.00

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under CAT.POL.A.330 for Class B aircraft, the landing distance from 50 ft must be multiplied by 1.43 (equivalent to using no more than 70% of the available runway).

Q2: At what airspeed does a piston-propeller aircraft achieve MAXIMUM ENDURANCE (maximum time in the ai

- [A]: At the speed for minimum power required ($V_{mp} \sim 0.76 V_{md}$)
- [B]: At the speed for minimum total drag (V_{md})
- [C]: At V_{no}
- [D]: At stall speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. Piston engine fuel burn is proportional to brake horsepower. Flying at the bottom of the Power Required curve (V_{mp}) burns the minimum fuel per hour, maximizing endurance.

Q3: How does a DOWNHILL runway slope affect landing distance?

- [A]: Decreases landing distance
- [B]: INCREASES landing distance (approx +5% per 1% of downhill slope)
- [C]: Has zero effect on landing roll

- [D]: Increases tire wear only

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A downhill slope reduces the deceleration component of gravity, requiring more wheel braking and lengthening landing distance.

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 5: Class A Take-Off Speeds (CS-25 Hierarchy: V1, Vr, V2)

Syllabus Source: CAE Oxford (pp. 149-194) | Pass Target: >= 90% AviationExam

1. THE CS-25 TAKE-OFF SPEEDS SPECTRUM

Class A covers commercial jet transports certified under CS-25. Take-off is governed by an interlocking hierarchy of calibrated airspeeds:

Speed Code & Name	Aerodynamic & Regulatory Definition	Mandatory Legal Constraint / Relationship
Speed Code & Name	Aerodynamic & Regulatory Definition	Mandatory Legal Constraint / Relationship
Vs1g / Vsr (Reference Stall)	Stall speed at 1g normal acceleration in take-off configuration.	Baseline reference for all minimum safety speeds.
Vmcg (Min Control Ground)	Minimum speed to maintain directional control on runway using RUDDER ONLY following engine failure.	$V_{mcg} \leq V1$! (If engine fails below V_{mcg} , rudder cannot keep plane on runway!).
V1 (Decision Speed)	Maximum speed to abort take-off; minimum speed to continue take-off safely following engine failure.	$V_{mcg} \leq V1 \leq V_r$ and $V1 \leq V_{MBE}$ (Max Brake Energy Speed).
Vr (Rotation Speed)	Speed at which pilot initiates backward pull on control column to pitch up.	$V_r \geq V1$ and $V_r \geq 1.05 \times V_{mca}$.
Vmu (Min Unstick Speed)	Minimum speed at which aircraft can safely lift off ground without tail strike.	Demonstrated during flight test with tail dragging on runway.
Vlof (Lift-Off Speed)	Actual speed at which the main tires leave the runway surface.	$V_{lof} \geq 1.10 \times V_{mu}$ (all engines) or $\geq 1.05 \times V_{mu}$ (OEI).
V2 (Take-off Safety Speed)	Target climb speed reached at 35 ft screen height with one engine inoperative.	$V2 \geq 1.13 \times V_{sr}$ and $V2 \geq 1.10 \times V_{mca}$!

2. THE GOLDEN HIERARCHY OF TAKE-OFF SPEEDS

The strict sequential order of speeds during an accelerating take-off roll is:

- The Take-off Sequence: $V_{mcg} \leq V1 \leq V_r < V_{lof} < V2$. (Never can $V1$ be less than V_{mcg} , and never can $V1$ exceed V_r !).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Class A Jet Takeoff Speed Chain

Master Speed Chain: $V_{MCG} \leq V_{EF} < V1 \leq V_r \leq V_{LOF} < V2$. - V_{MCG} : Minimum control speed on ground using aerodynamic rudder only. - $V1$: Takeoff decision speed. - V_r : Rotation speed ($V1 \leq V_r$). - $V2$: Takeoff safety speed at 35 ft screen height ($V2 \geq 1.13 \times V_{SR}$). $V1$ Action Rule: 1. Prior to $V1$: ABORT TAKEOFF immediately for engine failure, fire, or major system failure. 2. At or above $V1$: CONTINUE TAKEOFF into the air, even with total engine failure! Stopping past $V1$ leads to catastrophic runway excursion.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Checking Take-Off Speeds Legality

SCENARIO (Classic EASA Flight Deck Trap Drill): A dispatch calculation for a Boeing 737 yields the following calculated speeds: - $V_{mcg} = 112 \text{ kt}$ - $V_{mca} = 118 \text{ kt}$ - $V_1 = 110 \text{ kt}$ - $V_r = 122 \text{ kt}$ - $V_2 = 130 \text{ kt}$ QUESTION: Is this speed schedule legal for departure under CS-25 regulations? If not, what must be corrected? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Check Constraint 1: $V_{mcg} \leq V_1$: - $V_{mcg} = 112 \text{ kt}$ - $V_1 = 110 \text{ kt}$ - Check: Is $112 \leq 110$? NO! V_1 (110 kt) is LESS than V_{mcg} (112 kt)! - WHY THIS IS DANGEROUS FOR DUMMIES: If an engine fails at 111 kt, you are above V_1 (committing to fly), but below V_{mcg} (meaning the rudder cannot keep the aircraft straight on the runway, causing a runway excursion!). - Mandatory Fix: V_1 MUST be raised to at least 112 kt ($V_1 \geq V_{mcg}$)! Step 2: Check Constraint 2: $V_r \geq 1.05 \times V_{mca}$: - Minimum allowable $V_r = 1.05 \times 118 \text{ kt} = 123.9 \text{ kt}$ (~124 kt). - Actual proposed $V_r = 122 \text{ kt}$. - Check: Is $122 \geq 124$? NO! V_r is 2 knots too low! - Mandatory Fix: V_r must be raised to at least 124 kt! FINAL ANSWER: The schedule is ILLEGAL on two counts! V_1 must be increased to at least 112 kt, and V_r must be increased to at least 124 kt.

[!] EXAM TRAP (AVIATIONEXAM) - The V_1 Decision Point Trap

- V_1 is the DECISION SPEED: The action to abort must be INITIATED BY V_1 ! - In modern regulations, V_1 is the MAXIMUM speed by which the pilot must have made the decision AND applied the first stopping action (brakes/spoilers). If an engine fails after V_1 , YOU MUST CONTINUE THE TAKE-OFF!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: Under CS-25 regulations, what is the strict relationship between V_1 (Decision Speed) and V_{mcg} (Minimum**

- [A]: V_1 must be less than V_{mcg}
- [B]: V_1 must be GREATER THAN OR EQUAL TO V_{mcg} ($V_1 \geq V_{mcg}$)
- [C]: V_1 is always 1.2 times V_{mcg}
- [D]: V_{mcg} only applies in flight

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. If V_1 were below V_{mcg} , an engine failure between V_1 and V_{mcg} would require continuing the take-off, yet the aerodynamic rudder force would be insufficient to prevent the aircraft veering off the runway.

Q2: Under CS-25, what is the minimum legal ratio for Take-Off Safety Speed (V_2) relative to reference stall speed

- [A]: $V_2 \geq 1.13 \times V_{sr}$
- [B]: $V_2 \geq 1.05 \times V_{sr}$
- [C]: $V_2 = V_{sr}$
- [D]: $V_2 \geq 1.40 \times V_{sr}$

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. Under CS-25.107, V_2 must provide a stall margin of at least 13% above reference stall speed: $V_2 \geq 1.13 V_{sr}$ (and at least $1.10 V_{mca}$).

Q3: If an engine fails at a speed HIGHER than V_1 on a commercial airliner take-off, what is the legally mandated

- [A]: Immediately apply maximum wheel braking and reverse thrust
- [B]: CONTINUE the take-off, rotate at V_r , and climb away at V_2

- [C]: Extend full flaps
- [D]: Cut the remaining engine

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Above V1, remaining runway is insufficient to stop the aircraft without overrunning. The aircraft is certified to safely climb away on the remaining engine(s).

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 6: Class A Take-Off Distances & Balanced Field Length

Syllabus Source: CAE Oxford (pp. 195-238) | Pass Target: >= 90% AviationExam

1. DECLARED DISTANCES ARCHITECTURE

Airport authorities declare four official runway distances that establish the take-off and landing boundaries:

Declared Distance Code	Physical Runway Components Included	Regulatory Meaning & Usage
Declared Distance Code	Physical Runway Components Included	Regulatory Meaning & Usage
TORA (Take-Off Run Available)	Paved physical runway length suitable for normal ground run.	Baseline distance for all wheel acceleration.
TODA (Take-Off Distance Available)	TORA + Clearway (unobstructed water/grass area, max 50% TORA).	Total distance to accelerate and climb to 35 ft screen height.
ASDA (Acc-Stop Distance Available)	TORA + Stopway (load-bearing overrun strip suitable for abort).	Total distance available to accelerate to V1 and stop completely.
LDA (Landing Distance Available)	Paved runway from threshold to end of roll.	Distance available for landing touch down and rollout.

2. THE BALANCED FIELD LENGTH CONCEPT

A runway is considered a 'Balanced Field' when: TODA equals ASDA (Clearway = Stopway).

- Balanced V1: The unique decision speed where Accelerate-Stop Distance (reject at V1) EXACTLY EQUALS the One-Engine-Inoperative Take-off Distance (continue at V1 to 35 ft screen height)!
- Variables that SHIFT V1: Higher Mass -> Increases V1. Higher Flap Setting -> Decreases V1. Uphill Slope -> Increases V1. Headwind -> Decreases groundspeed at V1. Contaminated Runway (water/slush) -> DRATICALLY REDUCES V1 (because braking friction is degraded, you must abort earlier)!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Declared Distances & Clearway L

Declared Distance Formulas: - TORA = Take-Off Run Available (paved surface). - ASDA = TORA + Stopway. - TODA = TORA + Clearway (legal limit: clearway cannot exceed 50% of TORA!). Balanced Field Length Concept: - Occurs when Accelerate-Stop Distance Required equals Accelerate-Go Distance Required with engine failure at V1. - Balances field length to minimize required runway and maximize allowable takeoff mass.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Declared Distances & Clearway Limits

SCENARIO (Standard EASA Airport Planning Drill): An airport publishes the following physical runway survey data: - Paved Runway length = 2,400 meters - Stopway paved overrun strip = 150 meters - Clearway unobstructed obstacle-free area = 1,400 meters

QUESTION: What are the legal declared distances for TORA, ASDA, and TODA under EASA CS-25 rules? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate TORA (Take-Off Run Available): - TORA is strictly the physical paved runway length = 2,400 meters. Step 2: Calculate ASDA (Accelerate-Stop Distance Available): - ASDA = TORA + Stopway - ASDA = 2,400 m + 150 m = 2,550 meters. Step 3: Calculate TODA (Take-Off Distance Available) with Clearway Limit Check: - Rule: Clearway credit CANNOT EXCEED 50% of TORA! - Maximum allowable clearway = $0.50 \times 2,400 \text{ m} = 1,200 \text{ meters}$. - The physical clearway is 1,400 m, but we can ONLY LEGALLY CREDIT 1,200 meters! - Legal TODA = TORA + Allowed Clearway = 2,400 m + 1,200 m = 3,600 meters! FINAL ANSWER: TORA = 2,400 m; ASDA = 2,550 m; TODA = 3,600 m (limited by the 50% clearway rule!).

[!] EXAM TRAP (AVIATIONEXAM) - The Clearway 50% TORA Rule Trap

- Even if an airport has 3 kilometers of open sea beyond the runway, you CANNOT add more than 50% of TORA to TODA! - E.g. If TORA = 2,000 m and Clearway = 1,500 m, TODA is NOT 3,500 m! TODA is capped at $2,000 + 1,000 = 3,000 \text{ m}$!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the maximum clearway length that can legally be included in Take-Off Distance Available (TODA)?**

- [A]: Unlimited
- [B]: Not more than 50% of TORA
- [C]: Equal to the stopway length
- [D]: 1,000 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under CS-25 and ICAO Annex 14, the length of clearway that can be credited to determine TODA must not exceed half the length of TORA.

Q2: In a 'Balanced Field Length' take-off calculation, which two distances are equal?

- [A]: TORA and LDA
- [B]: The Accelerate-Stop Distance and the One-Engine-Inoperative Take-Off Distance to 35 ft
- [C]: Ground roll and climb distance
- [D]: Clearway and runway length

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A balanced field exists when the distance required to accelerate to V1 and stop equals the distance required to accelerate, lose an engine at V1, and reach screen height.

Q3: How does CONTAMINATION (e.g. 5 mm standing water) affect the optimal V1 decision speed?

- [A]: Increases V1

- [B]: SUBSTANTIALLY DECREASES V1, because degraded wheel braking requires aborting earlier, and remaining runway allows continued take-off to a reduced 15 ft screen height
- [C]: Has zero effect
- [D]: Eliminates V1

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. On contaminated runways, braking friction is severely reduced. V1 must be reduced to allow stopping within ASDA if an abort occurs; screen height for continue is lowered to 15 ft.

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 7: Class A Take-Off Flight Path & 4 Climb Segments

Syllabus Source: CAE Oxford (pp. 239-282) | Pass Target: >= 90% AviationExam

1. THE NET FLIGHT PATH & THE 4 CLIMB SEGMENTS

The take-off flight path begins at 35 ft screen height (dry) or 15 ft (wet) with one engine inoperative, continuing to 1,500 ft AGL. It is divided into four distinct aerodynamic segments:

Climb Segment	Physical Boundaries & Speeds	Aircraft Configuration & Min Gross Gradient (Tw
Climb Segment	Physical Boundaries & Speeds	Aircraft Configuration & Min Gross Gradient (Twins)
1st Segment	35 ft screen height to Gear Retraction fully complete. Speed = V2.	Take-off flaps, gear extending/retracting, Take-Off Thrust (TOGA). Min Gross Gradient: Positive (>= 0.0%).
2nd Segment (CRITICAL!)	Gear Retracted up to Acceleration Height (min 400 ft AGL). Speed = V2.	Take-off flaps, gear UP, TOGA thrust. Min Gross Gradient: 2.4% for twins (3.0% for quads) -- THE MOST STRINGENT!
3rd Segment (Acceleration)	Level acceleration flight at min 400 ft AGL. Speed accelerates from V2 to V_FTO.	Flaps retracting sequentially, gear UP, TOGA thrust. Min Gross Gradient: Acceleration equivalent.
4th Segment (Final)	From Acceleration Height to min 1,500 ft AGL. Speed = V_FTO (Final Take-Off Speed).	Clean configuration (flaps UP), gear UP, Max Continuous Thrust (MCT). Min Gross Gradient: 1.2% for twins.

2. GROSS VS NET TAKE-OFF FLIGHT PATH

Obstacle clearance calculations are NOT based on actual test flight performance (Gross):

- Gross Flight Path: The actual climb performance achieved by a skilled test pilot in a brand-new aircraft.
- Net Flight Path (EASA Deduction): Gross flight path reduced by a mandatory regulatory penalty to account for engine wear and pilot technique: Deduct 0.8% for twin-engine jets! (Deduct 0.9% for 3-engines, 1.0% for 4-engines).
- Obstacle Clearance Margins: The Net Flight Path must clear all ground obstacles by at least: 35 ft vertically in straight flight, or 50 ft vertically during turns (max bank angle 15°)! The wings must clear terrain horizontally by at least 90 m (or 300 m if radar tracking unavailable).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Takeoff Climb Segments & Net Flight Path

Four Climb Segments (Engine Inoperative): - 1st Segment: 35 ft to gear retracted. Positive gradient (>0%). - 2nd Segment (Critical): Gear up, takeoff flaps, V2 speed. Min Gross Gradient: 2.4% (twins), 2.7% (tri-jets), 3.0% (quads). - 3rd Segment: Horizontal acceleration and flap retraction at 400 ft minimum. - 4th Segment: Clean configuration, MCT thrust up to 1,500 ft. Min Gross: 1.2% (twins). Net Flight Path: Gross gradient minus 0.8% (twins). Must clear all obstacles by at least 35 ft.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Net Flight Path Obstacle Clearance

SCENARIO (Classic CS-25 Obstacle Clearance Problem): A twin-engine jet takes off with an engine failure at V1: - Screen height (35 ft = 10.7 m) reached at end of TODA - In the 2nd segment, the aircraft achieves a Gross Climb Gradient of 2.8% - An obstacle of height 180 ft (55.0 m) is located 3,000 meters beyond the end of TODA QUESTION: What is the NET climb gradient, and what is the aircraft's NET clearance over the obstacle? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the NET Climb Gradient (Deduct 0.8% for twins!): - Gross Gradient = 2.8% - Net Penalty (CS-25 twin) = -0.8% - Net Gradient = 2.8% - 0.8% = 2.0% (0.020). Step 2: Calculate the Net Altitude Gained over the 3,000 m distance to obstacle: - Net Altitude Gained = Horizontal Distance x Net Gradient - Net Altitude Gained = 3,000 m x 0.020 = 60.0 meters. Step 3: Calculate the Net Aircraft Height at the obstacle position: - Net Height = Initial Screen Height (10.7 m) + Net Altitude Gained (60.0 m) - Net Height = 10.7 m + 60.0 m = 70.7 meters (approx 232 ft). Step 4: Check Clearance Margin above the Obstacle: - Obstacle Height = 55.0 meters (180 ft) - Net Clearance = Net Aircraft Height - Obstacle Height - Net Clearance = 70.7 m - 55.0 m = 15.7 meters (51.5 ft)! Step 5: Verify Legal Margin (Must be $\geq$ 35 ft): - Since 51.5 ft $>$ 35 ft, the aircraft legally clears the obstacle! FINAL ANSWER: Net Climb Gradient = 2.0%. Net Clearance = 15.7 m (51.5 ft), which comfortably exceeds the 35 ft legal requirement.

[!] EXAM TRAP (AVIATIONEXAM) - The 2nd Segment Gradient Trap

- The 2nd segment is the MOST CRITICAL climb segment in ATPL exams! - Twin-engine minimum gross gradient = 2.4% (Net = 1.6%). - If high elevation or hot temperatures degrade climb below 2.4%, Take-Off Mass MUST be reduced, even on an infinitely long runway!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under CS-25, what is the mandatory Gross to Net gradient deduction for a TWIN-ENGINE transport aircraft?

- [A]: 0.5%
- [B]: 0.8% gradient reduction
- [C]: 1.0%
- [D]: 1.5%

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. CS-25 mandates a gradient penalty of 0.8% for 2-engine aircraft, 0.9% for 3-engine aircraft, and 1.0% for 4-engine aircraft to convert gross flight paths into legal net flight paths.

Q2: What is the minimum required GROSS climb gradient in the SECOND SEGMENT for a twin-turbofan aircraft?

- [A]: 0.0%
- [B]: 2.4% gross climb gradient
- [C]: 1.2%
- [D]: 3.0%

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. In the second segment (gear up, flaps take-off, V2, TOGA thrust), a twin must achieve at least a 2.4% gross climb gradient (yielding a 1.6% net gradient).

Q3: What is the minimum vertical obstacle clearance required between the NET take-off flight path and an obstacle?

- [A]: 50 ft
- [B]: 35 ft vertical clearance
- [C]: 100 ft
- [D]: Zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. CS-25 requires the net take-off flight path to clear all obstacles by at least 35 ft vertically in wings-level flight (increased to 50 ft during turns).

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 8: Class A En-Route Engine-Out Drift Down & Obstacles

Syllabus Source: CAE Oxford (pp. 283-316) | Pass Target: $\geq 90\%$ AviationExam

1. THE ENGINE FAILURE IN CRUISE PHYSICS

When an engine fails during cruise at high altitude (e.g. FL 370), the aircraft cannot maintain altitude because drag at that level exceeds maximum continuous thrust (MCT) of the remaining engine:

- **Drift Down Procedure:** The aircraft trades potential energy (altitude) for kinetic energy (airspeed), descending gradually at optimal single-engine speed (V_{md} / LRC speed) to maximize distance flown during descent toward the stabilizing ceiling.
- **Stabilizing Altitude (OEI Ceiling):** The level altitude where remaining engine thrust at MCT exactly balances aircraft drag. As fuel burns off and the aircraft gets lighter, the stabilizing ceiling gradually climbs!
- **Net En-Route Flight Path (1.1% Deduction):** Under CS-25 for twin-engine aircraft, the net en-route flight path is derived by deducting a 1.1% climb gradient penalty from gross data (1.4% for 3-engine, 1.6% for 4-engine).

2. EN-ROUTE OBSTACLE CLEARANCE REQUIREMENTS

Under EASA AIR-OPS (CAT.POL.A.215), the one-engine-inoperative net drift-down flight path must clear all high terrain and obstacles within 10 NM of the route centerline by AT LEAST 2,000 FT VERTICALLY!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): En-Route Driftdown Obstacle Clearance

Driftdown Procedure: Step 1: Set MCT (Maximum Continuous Thrust) on live engine. Step 2: Pitch to driftdown speed (approx L/D max) and disconnect altitude hold. Step 3: Turn 90 degrees off airway centerline per ATC oceanic/enroute contingency. Step 4: EASA Obstacle Margin: Net driftdown path must clear all terrain within 10 NM of track by at least 2,000 ft! Step 5: If single-engine ceiling cannot clear terrain, designate pre-computed escape routes to divert fields.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Verifying Drift Down Terrain Clearance

SCENARIO (Alpine Mountain Crossing Drift Down Drill): A twin-jet airliner is cruising at FL 350 over the Alps: - An engine fails at point A - The OEI drift-down table shows that the net flight path passes through 14,500 ft MSL at the critical ridge - The highest mountain peak within 10 NM of track has an elevation of 12,200 ft

MSL QUESTION: Is the drift-down flight path legal under EASA CAT.POL.A.215 regulations?

SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Identify the mandatory EASA

terrain clearance margin: - Required Vertical Clearance = 2,000 ft above terrain

within 10 NM of track. Step 2: Calculate the Minimum Allowable Net Flight Path

Altitude: - Min Net Altitude = Peak Elevation + Required Clearance - Min Net

Altitude = 12,200 ft + 2,000 ft = 14,200 ft MSL. Step 3: Compare Actual Net Altitude

with Minimum Required Altitude: - Aircraft Net Altitude at Ridge = 14,500 ft MSL -

Minimum Required Altitude = 14,200 ft MSL - Actual Vertical Margin = 14,500 ft -

12,200 ft = 2,300 ft! Step 4: Check Compliance: - 2,300 ft > 2,000 ft legal

requirement -> COMPLIANT! FINAL ANSWER: Yes, the flight path is legal. The aircraft

clears the peak with 2,300 ft net margin, exceeding the 2,000 ft regulatory

requirement.

[!] EXAM TRAP (AVIATIONEXAM) - The 10 NM Corridor & 2,000 ft Margin Trap

- Obstacle search corridor width is 10 NM ON EITHER SIDE of track (20 NM total

width). - Clearance requirement is 2,000 ft (NOT 1,000 ft!). 1,000 ft is only

permitted if the aircraft has reached its level stabilizing altitude and navigational

accuracy is within 95%.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA regulations, by what vertical margin must the OEI NET flight path clear all obstacles within 10 NM

- [A]: 500 ft
- [B]: At least 2,000 ft vertically
- [C]: 1,000 ft
- [D]: 35 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. CAT.POL.A.215 requires the one-engine-inoperative net flight path to have a positive slope at least 2,000 ft above all terrain within 10 NM on each side of the intended track.

Q2: What is the regulatory gradient penalty applied to gross drift down performance to obtain the NET drift down

- [A]: 0.5%
- [B]: 1.1% gradient deduction
- [C]: 2.0%
- [D]: Zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. CS-25 requires deducting 1.1% from gross climb/descent gradient data for twin-engine aircraft during the en-route phase.

Q3: What speed should the flight crew fly to maximize distance traveled during an engine-out drift down?

- [A]: Maximum operating Mach (Mmo)
- [B]: The single-engine drift down speed (approx Vmd / Long Range Cruise speed)
- [C]: Stall speed
- [D]: Minimum clean speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Flying at the speed for minimum drag (Vmd) yields the maximum lift-to-drag ratio (L/D max), resulting in the shallowest glide angle and maximum distance to the OEI ceiling.

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 9: Class A Landing Performance & Factored Distance

Syllabus Source: CAE Oxford (pp. 317-360) | Pass Target: $\geq 90\%$ AviationExam

1. CLASS A LANDING REQUIREMENTS & SCREEN HEIGHT

Class A landing performance begins at a screen height of 50 ft above the runway threshold at target approach speed V_{ref} :

- Reference Landing Speed (V_{ref}): Target speed over threshold: V_{ref} MUST BE AT LEAST $1.23 \times V_{sr0}$ (stall speed in landing configuration).
- Landing Distance Available (LDA): The physical paved runway declared available from threshold to end of roll.
- The 60% Rule (Dry Runway Factor $\times 1.67$): Under EASA CAT.POL.A.230, a turbojet aircraft must be capable of landing from 50 ft to a complete stop within 60% OF THE LDA! Therefore:
Required Factored Landing Distance = Actual Measured Distance / 0.60 = Actual Distance $\times 1.67$.
- Wet Runway Factor ($\times 1.15$ of Dry = $\times 1.92$ Total!): If the runway is forecast to be wet or slippery at ETA, the required landing distance is 115% OF THE FACTORED DRY DISTANCE:
Required Wet Landing Distance = Actual Distance $\times 1.67 \times 1.15$ = Actual Distance $\times 1.92$!

2. REVERSE THRUST REGULATORY CREDIT

Under CS-25 certification for dispatch purposes, NO CREDIT is permitted for thrust reversers on dry or wet runways! The aircraft must be capable of stopping using wheel brakes and aerodynamic spoilers alone. (Reverse thrust credit is only considered in non-normal in-flight assessments or contaminated runway models).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Class A Jet Landing Factored Dis

Rules for Commercial Jet Dispatch: Step 1: Measured AFM landing distance from 50 ft threshold to full stop. Step 2: Dry Runway Factor = $\times 1.67$ (Must stop within 60% of LDA: $1 / 0.60 = 1.67$). Step 3: Wet Runway Factor = Dry Factored Distance $\times 1.15$. Step 4: Crucial Exam Rule: Reverse thrust credit is FORBIDDEN in dry dispatch landing calculations! Spoilers and wheel brakes only.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Dispatch Landing Distance Check Dry vs Wet

SCENARIO (Commercial Airline Dispatch Drill): A Boeing 737 is dispatching to a destination with a Landing Distance Available (LDA) of 2,200 meters: - Flight Manual Unfactored Landing Distance from 50 ft = 1,200 meters - Destination forecast: Rain showers at ETA (Runway WET!) QUESTION: What is the Required Dry Landing Distance? What is the Required Wet Landing Distance? Can the aircraft legally dispatch? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate Required DRY Landing Distance (The 60% Rule = $\times 1.67$): - Required Dry Distance = Actual Distance / 0.60 = 1,200 m / 0.60 = 2,000 meters. - (On a dry runway, 2,000 m $\leq$ 2,200 m LDA -> Legal for dry runway). Step 2: Calculate Required WET Landing Distance (Apply 115% factor to Dry): - Required Wet Distance = Required Dry Distance $\times 1.15$ - Required Wet Distance = 2,000 m $\times 1.15$ = 2,300 meters! - (Notice: 1,200 m $\times 1.92$ = 2,304 meters). Step 3: Compare Required Wet Distance with Runway LDA: - Required Wet Distance = 2,300 meters - Available LDA = 2,200 meters - Verdict: 2,300 m $>$ 2,200 m -> EXCEEDS AVAILABLE RUNWAY BY 100 METERS! FINAL ANSWER: The aircraft CANNOT legally dispatch! Landing mass must be reduced, or an alternate destination selected.

[!] EXAM TRAP (AVIATIONEXAM) - The 60% Rule vs 1.67 Factor Distinction Trap

- Stating 'Actual distance must not exceed 60% of LDA' is mathematically identical to stating 'Required LDA must be at least Actual distance multiplied by 1.67 ($1 / 0.60 = 1.667$)'! - For Wet runways, multiply by 1.92 total! Always check whether the exam question asks for DRY or WET!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA CS-25, what is the minimum approach speed (V_{ref}) over the runway threshold for a commercial aircraft?

- [A]: $V_{ref} \geq 1.23 \times V_{sr0}$ (stall speed in landing configuration)
- [B]: $V_{ref} = V_{sr0}$
- [C]: $V_{ref} = 1.10 \times V_{mca}$
- [D]: Exactly 140 kt

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. CS-25.125 requires that V_{ref} be at least 23% above the reference stall speed in the landing configuration: $V_{ref} \geq 1.23 V_{sr0}$.

Q2: For a commercial turbojet landing on a DRY runway, what proportion of the Landing Distance Available (LDA) must be used?

- [A]: Exactly 50%
- [B]: Not more than 60% of the LDA (factoring distance by 1.67)
- [C]: 70%
- [D]: 100%

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under CAT.POL.A.230, the landing distance from 50 ft must not exceed 60% of LDA, ensuring a 40% runway safety buffer.

Q3: When calculating dispatch landing performance on a WET runway under CS-25, what additional factor is applied to the dry landing distance?

- [A]: 10%
- [B]: An additional 15% (giving a total factor of 1.92 times actual distance)

- [C]: 30%
- [D]: Zero factor

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A wet runway requires a 15% increase over the dry factored distance (Factored Dry x 1.15), reflecting reduced wet tire friction.

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 10: Reduced & Derated Take-Off Thrust (Flex Temp)

Syllabus Source: CAE Oxford (pp. 361-402) | Pass Target: >= 90% AviationExam

1. WHY USE REDUCED THRUST?

Jet engines deteriorate primarily due to high Turbine Gas Temperature (EGT) during take-off at full rated power. When an aircraft is light or runway is long, taking off with reduced thrust saves substantial engine overhaul costs and increases reliability without compromising safety!

2. THE ASSUMED TEMPERATURE METHOD (FLEX TEMP)

Modern turbofans are 'flat-rated' up to a corner point temperature (T_ref, typically ISA + 15°C / ~30°C). Above T_ref, engine thrust drops automatically to protect turbine blades from over-temperature:

- The Flex Principle: The pilot enters an artificial, fictitiously HIGH outside temperature (e.g. Assumed Temp = +52°C, when actual OAT is +20°C). The FADEC commands reduced take-off N1/EPR matching performance on a hot day!
- The 25% Maximum Reduction Limit: REGULATORY LIMIT: Reduced thrust MUST NOT BE LESS than 75% of maximum rated take-off thrust! (Maximum allowable thrust reduction is 25%!).
- Assumed Temp Prohibitions: The Assumed Temperature Method is STRICTLY FORBIDDEN: (1) On contaminated runways (standing water, slush, ice), (2) When antiskid is inoperative, (3) When windshear is reported or suspected!

3. ASSUMED TEMPERATURE VS FIXED DERATED THRUST

Thrust Reduction Method	Certification Status & Minimum Control Speeds	Pilot Operating Authority in Cockpit
Thrust Reduction Method	Certification Status & Minimum Control Speeds	Pilot Operating Authority in Cockpit
Assumed Temperature (Flex)	Certified at full rated thrust. Minimum control speeds (Vmcg, Vmca) are based on FULL RATED THRUST.	Pilot MAY advance thrust levers to full TOGA at any time during take-off!
Fixed Derate (Derated Thrust)	Certified as a completely separate, lower-rated engine (e.g. 24k rating derated to 22k). Vmcg and Vmca are LOWER!	Pilot MUST NOT push levers to full TOGA below Vmca (doing so risks violent loss of directional control)!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Assumed / Flexible Temperature

Principles of Flex Takeoff: Step 1: Input fictitious higher temperature (T_flex > OAT) into FMS/MCDU. Step 2: FADEC meters less fuel, delivering reduced thrust to prolong engine turbine life. Regulatory Limits (EASA CAT.POL.A): - Maximum allowable thrust reduction: 25% below full rated takeoff thrust. - STRICTLY FORBIDDEN on contaminated runways (water >3 mm, slush, ice). - STRICTLY FORBIDDEN with anti-skid inoperative or reported windshear.

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Checking Assumed Temperature Feasibility

SCENARIO (Airline Dispatch Problem): An Airbus A320 is operating from a 3,500 m runway: - Actual Outside Air Temperature (OAT): +18°C - Flat-rated corner temperature (T_{ref}): +30°C - Performance tables indicate aircraft can take off safely at current mass up to an assumed temperature of +54°C - Full rated thrust produces 120 kN per engine - At +54°C assumed temperature, thrust output is 93 kN per engine QUESTION: Can this take-off be conducted legally using Assumed Temperature? What is the percentage thrust reduction? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Check Condition 1: Is Assumed Temperature HIGHER than actual OAT? - Assumed Temp (+54°C) > Actual OAT (+18°C) -> PASSED! - (Assumed temperature must ALWAYS be higher than actual OAT and higher than T_{ref}). Step 2: Check Condition 2: Calculate Percentage Thrust Reduction: - Full Rated Thrust = 120 kN - Reduced Thrust at +54°C = 93 kN - Actual Thrust Reduction = 120 kN - 93 kN = 27 kN. - Percentage Reduction = (27 kN / 120 kN) x 100 = 22.5% reduction! Step 3: Compare with the 25% Regulatory Limit: - Regulatory Maximum Reduction = 25.0% - Actual Reduction = 22.5% - Check: 22.5% ≤ 25.0% -> PASSED! The reduction is within the legal 25% limit! Step 4: Check Runway Condition: - Provided the runway is NOT contaminated and antiskid is operational, the takeoff is 100% legal! FINAL ANSWER: Yes! The take-off is legal with Flex Temp +54°C, yielding a 22.5% thrust reduction (below the 25% legal cap).

[!] EXAM TRAP (AVIATIONEXAM) - The Vmcg Difference: Flex vs Derate Trap

- Under ASSUMED TEMPERATURE: Vmcg is based on FULL RATED THRUST (high Vmcg). Moving thrust to full TOGA is safe. - Under FIXED DERATE: Vmcg is based on the REDUCED thrust (low Vmcg). Pushing full TOGA below the full-thrust Vmcg will cause uncontrollable rudder divergence off the runway!

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the maximum permissible thrust reduction when using the ASSUMED TEMPERATURE METHOD for

- [A]: 10%
- [B]: Not more than 25% of full rated take-off thrust
- [C]: 50%
- [D]: Unlimited

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under CS-25 regulations, thrust reduction using the assumed temperature method is capped at a maximum of 25% below full rated thrust (minimum 75% thrust remaining).

Q2: Under which of the following operational conditions is the use of the ASSUMED TEMPERATURE METHOD

- [A]: At sea level
- [B]: On CONTAMINATED runways (e.g. standing water, slush, or ice)
- [C]: At temperatures below ISA
- [D]: On runways longer than 3,000 m

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Assumed temperature take-off is banned on contaminated runways due to unpredictable tire friction and the requirement to maintain maximum acceleration and stopping margins.

Q3: What is the fundamental difference between 'Assumed Temperature' (Flex) and 'Derated Thrust' regarding

- [A]: There is no difference
- [B]: Assumed temperature uses Vmcg based on FULL RATED thrust; Derated thrust establishes a certified LOWER Vmcg corresponding to the lower derated thrust rating
- [C]: Derate increases Vmcg
- [D]: Flex has no Vmcg

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Fixed derate is certified at a lower thrust level, allowing lower minimum control speeds (Vmcg and Vmca), whereas assumed temperature retains the higher full-rated Vmcg.